

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1.

What type of evaluation consists of user studies and/or cognitive experiments?

1 / 1 point
- ☒ Human Evaluation

☐ Functional Evaluation

☐ Evaluation of Fairness and Bias

Correct

Correct! Human Evaluation approaches include user studies and cognitive experiments.

2.

What type of evaluation consists of synthetic (or simplified) tasks and benchmarking?

1 / 1 point

☐ Specialized Evaluation

☐ Functional Evaluation

☒ Proxy Tasks

Correct

Correct! Proxy tasks include synthetic or simplified tasks and benchmark datasets.

3.

When evaluating an XAI system, where should you start?

1 / 1 point

☒ Start with a comparison to baselines or alternative methods

☐ Start with human evaluation

☐ Start with a comparison to synthetic or simplified tasks

Correct

Correct! Comparing the performance and quality of explanations generated by different XAI techniques or against baseline methods should always be done to start your evaluation process.

4.

Which of the following is NOT a security concern in XAI systems?

1 / 1 point

☐ Adversarial Attacks

☐ Tampering or Manipulation

☐ Data privacy and protection

☒ Out-of-distribution robustness

Correct

Correct! Out of distribution robustness is a consideration of robustness in XAI systems rather than a security concern.

5.

Which of the following are adversarial attacks (check all that apply)?

1 / 1 point

☒ When an attacker crafts carefully designed input data to manipulate the model's output or the explanations provided by the XAI system

Correct

Correct! This is an example of an adversarial attack.

☐ When an attacker observes the model's predictions without interfering

☒ When an attacker attempts to extract or replicate the underlying ML model by probing the XAI system with carefully crafted inputs and analyzing the explanations.

Correct

Correct! This is an example of an adversarial attack.

6.

Match the definition to its term: "Explanations provided by XAI systems should accurately reflect the true reasoning behind the model's predictions":

1 / 1 point

☐ Model stability

☐ Out-of-distribution robustness

☒ Explanation Fidelity

Correct

Correct! The definition of explanation fidelity is in fact that explanations provided by XAI systems should accurately reflect the true reasoning behind the model's predictions.

7.

What is model regularization used for in the context of robust model training?

1 / 1 point

☒ Used to prevent overfitting and improve the generalization ability of the model

☐ Used to find model parameters that optimize performance under various perturbations or uncertainties

☐ Used to make the model more robust by learning to correctly classify these adversarial examples, thus improving its resilience against adversarial attacks

Correct

Correct! Model regularization is used to prevent overfitting and improve the generalization ability of the model. By reducing overfitting, the XAI system becomes more stable and less sensitive to small perturbations in the input data, improving its robustness.

8.

What is the main difference between watermarking and fingerprinting?

1 / 1 point

☐ Watermarks are often visible or easily extractable from the explanation while fingerprints are hidden.

☒ Fingerprints are often visible or easily extractable from the explanation while watermarks are hidden.

☐ Both watermarks and fingerprints are visible, but only fingerprints are easily extractable from the explanation.

Correct

Correct! Unlike watermarks, fingerprints are often visible or easily extractable from the explanation.

9.

To identify potential vulnerabilities, weaknesses, or inconsistencies in the XAI system, it is important to (check all that apply):

1 / 1 point

☒ Perform rigorous testing and evaluation

Correct

Correct! Rigorous testing and evaluation on diverse datasets, including out-of-distribution and adversarial examples should be used to identify potential vulnerabilities, weaknesses, or inconsistencies in the XAI system.

☒ Quantify and monitor the fidelity of explanations

Correct

Correct! Quantifying and monitoring the fidelity of explanations should be done using a sensitivity analysis or perturbation-based evaluation to identify potential vulnerabilities, weaknesses, or inconsistencies in the XAI system.

☒ Conduct user studies

Correct

Correct! Conduct user studies to evaluate the interpretability and usefulness of the explanations and help identify potential vulnerabilities, weaknesses, or inconsistencies in the XAI system.

10.

Guidance to users on the limitations, potential biases, and proper interpretation of explanations can help foster this Culture of _____.

1 / 1 point

☐ Explainable Use

☒ Responsible Use

☐ Transparent Use

☐ Interpretable Use

Correct

Correct! Fostering a culture of responsible use and ethical decision-making based on the insights provided by the XAI system is important to both robustness and security. Guidance to users on the limitations, potential biases, and proper interpretation of explanations can help foster this culture of responsible use.